

SumoBot Team Orange

Final Report

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DISCLAIMER

This report was prepared by students as part of a university course requirement. While considerable effort has been put into the project, it is not the work of licensed engineers and has not undergone the extensive verification that is common in the profession. The information, data, conclusions, and content of this report should not be relied on or utilized without thorough, independent testing and verification. University faculty members may have been associated with this project as advisors, sponsors, or course instructors, but as such they are not responsible for the accuracy of results or conclusions.

ABSTRACT

Sumobot exists as a competition held during Oregon State Universities Engineering Expo that is meant to demonstrate the iterative design process by designing, wiring, and creating an individual robot to compete in simulated sumo matches. The format is made up of a circular 1.54 diameter mat called a “Dohyo” where participants would place their team’s robot in the ring and attempt to push the opponents out of it. Matches are held in 3 minute long rounds with a best of 3 format to allow time to prepare for competition. Robots are expected to be no bigger than 20.0 cm by 20.0 cm with a maximum weight limit of 3.0 kg.

Our team's design for the competition involves a rectangular body that houses 4 DC brushless motors within the frame. We decided on a 4 motor drive system because our team focused on having a high torque drive paired with a pivoting offensive mechanism in the front of our robot to bull doze opponents outside of the Doyle. For our electronics and control of our robot we decided on using a Battery Eliminator Circuit (BEC) due to its simple integration and preset software avoiding the need to code. Avoiding the need to code was preferred in our design decision as no team members had any practical experience coding and it allows more time to be dedicated towards other areas of our project like design and testing.

After the initial iterations of our robot, we tested it against the engineering requirements defined in the beginning of the term to determine if the robot has met the standards. From this, we would brainstorm, manufacture, and improve our robot to satisfy the unfulfilled requirements we determined from our test. We would repeat this process until all these requirements are satisfied then explored into the aesthetics of the robot: our final design achieved 8 out of the 10 engineering requirements.

ACKNOWLEDGEMENTS

We would like to use this page to acknowledge everyone that contributed to help bring this project to completion whom we could not have done without them. Those being Oregon State University School of Mechanical, Industrial, and Manufacturing Engineering (M.I.M.E) as well as our technical advisor and supervisor of this project Dr. Sarah Oman

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1 BACKGROUND

1.1 Introduction

Sumobot is an interactive competition where several teams are tasked with designing and building a robot from scratch to compete in the sumobot competition held during the OSU Engineering Expo. The match format is from 2-4 teams are placed onto a "Doyho" and then attempt to push the opponents outside of the ring. Our objective is to design and produce a robot that will be able to win the competition as well as providing an enjoyable experience for the expo attendees. Because our sponsor is Oregon State University, we would be providing a tangible benefit for the expo as we would be bringing in attendees who would then be interested in other projects presenting nearby

This project is important as it allows us to showcase our skills regarding our education at OSU as well as being able to demonstrate our prototyping and design process that made us land on our final design choice

1.2 Project Scope and Stakeholders

Sumobot is an interactive competition where several teams are tasked to design and build up a robot from scratch that are then placed onto a circular sumo mat and recreate sumo matches with up to 3 teams competing at the same time. The format is best of 3 matches and games have a maximum time limit of 3.0 minutes to encourage quick and enjoyable spectating from expo attendees. The competition focuses on creativity and designing in order to demonstrate the skills involved with rapid prototyping. Another goal is to be able to describe and demonstrate the wiring and controlling of the robot from outside of the ring. This also allows up to provide community outreach towards visiting students of spectators interested in STEM fields.

The people involved in this project are the 3 Mechanical Engineering student team members who are the core team members of this project, Aiden Aerni, Joshua Calderon, and Andrew Dillon. We are also joined by our project supervisor who also doubles as the representative for our primary stakeholder, Dr. Sarah Oman, as we are representing Oregon State University's school of Mechanical, Industrial, and Manufacturing Engineering (M.I.M.E). Secondary stakeholders for this project include the end users, specifically the spectators and attendees who will observe the Sumobot competition during the expo.

1.2.1 Risks and Assumptions

General Risks that would be associated with this project would involve scope creep due to the wide range of creativity that each team has been given as well as considering material strengths and weaknesses that might prove fragile in tested environments. External risks outside the process would involve the inability to source the materials required to complete the project due to supply chain and logistical constraints that could be caused by the increase in tariffs. Steps that could be taken to avoid the external risk factors would involve ordering all required components well in advance, ideally before we would need them to allow for a greater downtime to refine and perform testing on a working concept.

Assumptions that we would hold throughout the designing and testing process would involve specifying that the frame for our robot would be made out of 3D filament, which would have to be taken into account in order to see how it would impact the final design process. Another assumption that would be held is ordering parts for the continental United States, and not overseas, so we could avoid running into supply chain issues from foreign companies during the shipping process.

2 DESIGN PROCESS

To proceed with the design process, the team first created a customer requirements criteria for which the robot would be capable of achieving. The criteria was based on the following goals that the team has created: win as many Sumo matches as possible, design the robot to be aesthetically pleasing, and have an interesting feature about the robot that makes our robot unique. The criteria was also based on the team's sponsor preference and the chosen robot design strategy of creating a robot that sacrifices speed and maneuverability for a high torque drive. The justification behind the team's strategy was rooted in the idea that every Sumo match primarily consisted of "head to head" force battles. Having a high torque drive system would allow the robot to forcefully push opposing robots out of bounds.

Using the criteria, each team member separately brainstormed two concepts each that attempted to satisfy the criteria as best as it could. A datum design was also determined among all group members that acted as a benchmark. This benchmark would be used in the Decision Matrix when evaluating the developed concepts. The following descriptions and images describe the created concepts....

Idea 1 (Figure 1) - A low square frame that utilized a tank tread drive system. Tank threads were chosen to increase contact surface with the ground, increasing grip to avoid slipping when pushing opposing robots. A low square frame was chosen to maximize space for robot components and have a low center of gravity to avoid tipping

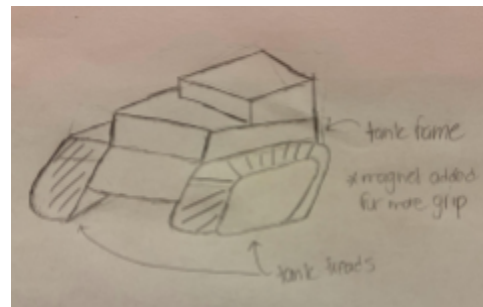


Figure 1.

Idea 2 (Figure 2) - A square frame that utilized a drive system that includes more than four wheels. Designing a drive system with this characteristic was chosen to increase contact surface and torque for pushing opponents. Uni directional (Omni/Meccanum) wheels would be used to allow for increased maneuverability around the field.

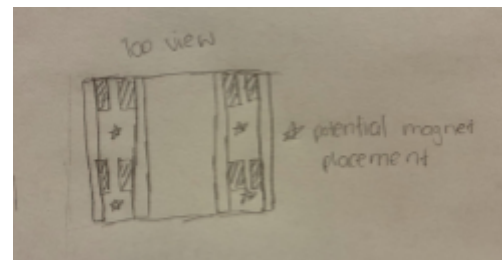


Figure 2.

Idea 3 (Figure 3) - A low swept shell frame with a tank thread drive system. Tank treads were chosen to increase contact surface with the ground. This would increase grip to avoid slipping while pushing opposing robots. A low shell frame design was chosen to lift opposing robot's wheels off the ground without tipping allowing for more leverage.

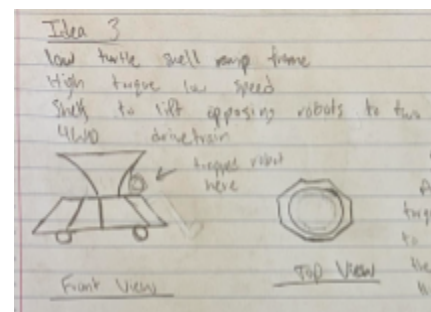


Figure 3.

Idea 4 (Figure 4) - Circular frame with tank thread drive system. Tank threads were chosen to increase contact surface with the ground. This would increase grip to avoid slipping while pushing opposing robotics. Exentable arms would be implemented to deploy after the match has been initiated to increase the overall effective area of the robot.

Idea 5 (Figure 5) - Low rectangular frame with tank thread drive system for a low center of gravity. Tank treads were chosen to increase contact surface with the ground, increasing grip to avoid slip when pushing opposing robotics. A suction emergency break would be implemented to act as an emergency break when necessary to avoid being pushed out of bounds.

Idea 6 (No picture) - Cylindrical frame with a drive system that focused on maneuverability. The concept would also include an emergency brake concept system with deployable arms. The idea was to “ground” the robot, then sweep around the field pushing opposing robots out of bounds.

Datum (No picture) - Low square frame with a four wheel drive system for a low center of gravity and stability. The drive system would have moderate torque and the frame would include magnets to increase the robot's downward force. This would compensate for the lower torque and allow for increased grip to avoid slipping when pushing opposing robots.

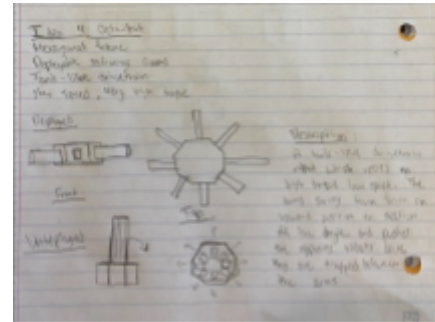


Figure 4.

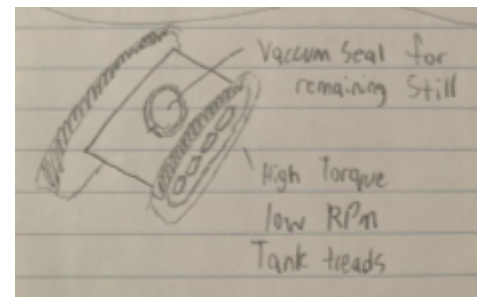


Figure 5.

After concepts were created, the team used a Decision Matrix to score each design and identify the final concept. Each concept was assessed in how it compared against the datum for each criteria: “+” means the concept performed better than the Datum for that criteria, while “-” means the team thought it would perform worse than the Datum for that criteria. The justification behind comparing each concept with respect to the datum is to ensure standardized evaluation between all of the concepts. Filling out the matrix, the team determined that Ideas three and four captured the criterias the best, and that a combination between the two would be the final concept that the team would pursue for the rest of the project.

	A	B	C	D	E	F	G	H	I
1	Ideas	Datum	Idea 1	Idea 2	Idea 3	Idea 4	Idea 5	Idea 6	
2	Criteria								
3	Robot can produce at least 8lbs of force		+	-	+	+	+	-	
4	Robot can move along the x and y plane easily		-	+	+	-	-	-	
5	Robot must be around 3kg +/- 0.1kg		+	+	+	+	+	+	
6	Robot must fit within a 20cm by 20cm box		+	+	+	+	+	+	
7	Robot top speed of 2 feet per second		-	+	-	-	-	-	
8	Robot must withstand a force of 8lbs without moving		+	-	+	+	+	-	
9	Robot must decelerate of at least a rate of 8ft/s ²		-	-	+	+	-	-	
10	Robot must use RC control for operation		+	+	+	+	+	+	
11	Robot part must be accessible within one day of shipping		-	+	+	-	-	-	
12	Robot build must include one form of offensive capability		-	-	+	+	-	+	
13									
14									
15	Sum of positives		5	5	6	7	7	5	4
16	Sum of negatives		5	5	4	3	3	5	6
17									
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Our group came to to conclusion that ideas 3 and 4 would fit our engineering specifications the best: low frame, prioritizing a high torque drive system, and having one offensive capability. From this we will start to to create low fidelity prototypes to design a robot that fit these concepts to further progress this project.

Figure 3. Decision matrix used to access concepts

The new final concept, a combination between Ideas three and four, was determined by the team to be a square frame; operate with high torque; use a four wheel tank tread drive system; with an offensive arm mechanism that would remove the front wheels of the opposing robot from contacting the ground. A square frame was chosen over other designs as it offers the most space to house components, while also providing the most stability during contact. The justification behind the tank treads instead of typical four wheel drive is to provide the robot with as much contact with the ground as possible. An article titled *Mini-Sumo Robot Design with Respect to the Control System* highlighted the importance of wheel selection for grip so the robot can ensure maximum pushing efficiency [1, pg. 5]. The team also determined that having an offensive arm to lift the front wheels of an opposing robot off the ground to be valuable as it lessens the grip of the opposing robot with the ground making it less challenging to push.

The Sumobot would be controlled using a radio control (RC) system from a Battery Eliminator Circuit (BEC). This BEC circuit system would allow all of the robot's systems to be powered from a singular battery compared to other circuits which require separate batteries for the transmitter [2]. This is important as having one battery significantly reduces the weight of the robot and would allow for more effortless integration of necessary components like the offensive mechanism. The team determined that the BEC circuit is also more ideal compared to other options such a microcontroller as it is easier to implement.

3 DESIGN PROPOSAL – First Term

Contained in this chapter is our initial design as well as several steps that we took in order to continuously develop the components throughout the first term of the capstone project. This will go over our initial frame design and how it has been changed in order to streamline the motor positioning

From our first term, we had a design process beginning with isolating and segmenting the different sections for our robot into 3 distinct and different areas to focus on, those being the drivetrain, body/internal components, and external offensive mechanism. From there, our team brainstormed different possible combinations for each core component with many similar results for different possible combinations. Some of those were generic box-shaped all the way to unique tread-like mechanism for our drivetrain with curved and movable mechanisms that could be added onto our robot in order to assist in the competition

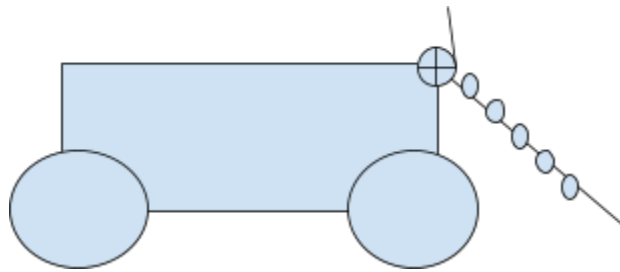


Figure 6: Initial Rough Design of robot

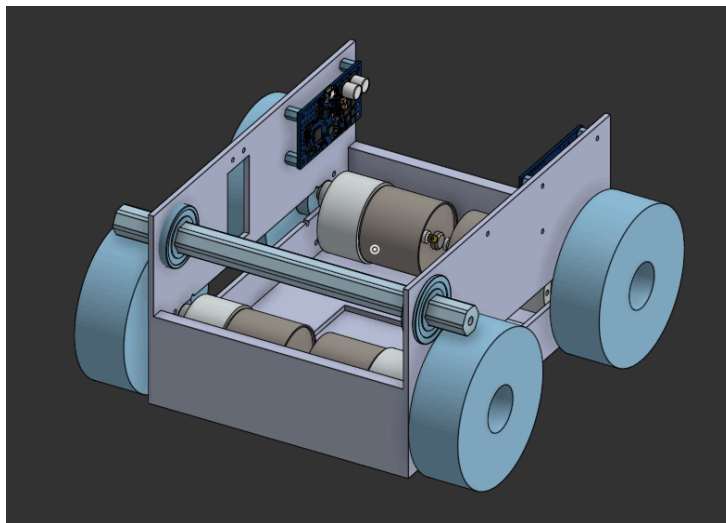


Figure 7 Final Design of Term 1 for the robot

From the various concept combinations, six final preliminary designs were developed. Four of these designs incorporated a tank-tread-inspired configuration, reflecting the team's initial preference for an omni-wheel-based drivetrain to maximize traction and maneuverability on the competition surface. However, the tank tread drivetrain was abandoned as it was considered too costly and would not provide as tangible of a benefit as we had hoped. Specifically with a bigger disadvantage when needing to turn, while that feature was considered a priority, we wanted to instead have 4 motor drivetrain connected to 4 wheels to more easily perform these operations.

Concerning the body/internal components, in order to streamline our design process, we decided on a standard rectangular body in order to house the different hardware and software that would need to be

assembled in our final design. [Fig. 8] For our battery, we picked a 2200 mAh battery to power our robot, which we felt would provide ample amps in order to operate all functions in our robot. For our motors we picked four 131:1 Metal Gearmotor 37Dx57L mm 12V motors. When considering the electrical components in our design, we selected on having 2 motor controllers, with 1 on each side that we could then plug in a power to the left and right sides individually, leading to easier control when in competition. However in our initial design selection, we originally were attempting to code the drivetrain and then use bluetooth to connect to a controller in order to operate it that way. This had us run into several issues with this option, some of those being that team members were not exactly confident in their coding skills to develop the drivetrain and have it operational within the specified deadline. Which is why we decided to switch our controlling mechanism to RC control, which would allow us to just need to wire the RC receiver into the robot, and connect it to the motor controllers to make it operational, allowing us to easily implement it into our design rather than have to develop a separate wiring drivetrain



Fig. 8: Initial Rectangular Frame with motor housing

For our secondary research into the offensive mechanism which we would then make it easily attachable and detachable for testing and prototyping onto the robot, many of the proposed mechanisms would be mainly split up into 2 separate distinctions.

a.) Rotatable arm

Our initial thoughts were something that we saw from a video Dr. Oman showed us during one of our scheduled meetings and having a motor controlled arm at the top of the robot spinning fast to knock their opponents quickly outside of the ring. We found that interesting and wanted to design our robot with a focus on high torque with an arm that would be deployed when the competition would start to give us a greater area to catch opponents and push them out of bounds. Within the ruleset of stating that there was no specified height limit, theoretically we could include a massive arm that would be deployed at the beginning of the game match, as that would fit within the competition ruleset. However since there is a set weight limit of 3.0 kg, that would involve dedicating valuable weight to a functioning arm that could rotate back and forth, which could hinder the robot by not having its center of mass low to the ground, leaving a risk of having an opponent easily push us outside the ring. However, this aspect of the offensive mechanism was eventually dropped due to if the arm were to be deployed, then we would suffer mobility operations of our robot as our frame would end up clipping the arm, causing it to get stuck

b.) Lifiable Shield

Our other mechanism was a shield that would drop down in front of our robot, ideally hooked up to servo motors or with a stop-gap that would prevent the shield from going back to its pre-dropped position. This attachment would have a slight dip within its shape to try and catch the opponent's front wheels and lift them up, such that they would lose significant traction on the remaining wheels that would allow our robot to easily beat it in a head-to-head pushing match. Another shield mechanism that was suggested was to attach rollers to a ramp that when an opponent would go up on the ramp, those rollers would catch its wheels, restricting its movement which would allow our robot to drive the opponent to the edge of the ring. With this design, we would also attach a lip at the top of the ramp in order to prevent the opposing robot from driving on the top roof of our robot, which could possibly damage the components within our robot.

The version of the liftable shield would eventually be what we decided on our offensive mechanism, with a ramp that would be held up and tilted to the back in order to satisfy the initial starting parameters of fitting into the 20.0 cm by 20.0 cm square. [Figure 9] When the match would begin, we would move backwards in order to flip it down so it is flush with the ground, while using a servo motor to push the ramp on a strictly linear path back and forth, while using springs attached from the shaft of the offensive mechanism, to the back of the robot. So when the ramp reaches max displacement, as the servo extends, it moves from the ramp, allowing the springs to pull it back to its original position. Allowing us to repeat that motion as many times as necessary. [See Appendix 8.1](#)

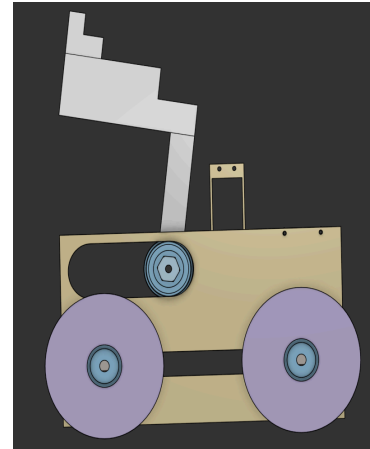


Figure 9: Lifted frame in pre-match position

Our results from the end of the first term to now involved keeping a similar frame, but instead there was a decision to remove most of the physical barriers in the initial frame that were supposed to hold most of the components in place, however there were some issues getting the wires and motors to be in their respective locations, leading to the axles being uneven, causing our design to drift. In order to rectify this, we then included a large oval throughout the left and right sides of the frame to have an easier time to slide the motors in and screw them to the frame, leading to the inner walls not being necessary in our design. Other changes were to adjust the motor controllers which were horizontal in our initial draft, to vertical in order to optimize the amount of space each motor controller would take up on either side of the inner walls of the robot. Leaving more room in order to have proper cable management as well as avoiding cramming components together in enclosed places, as that could damage them further. [Figure. 10]

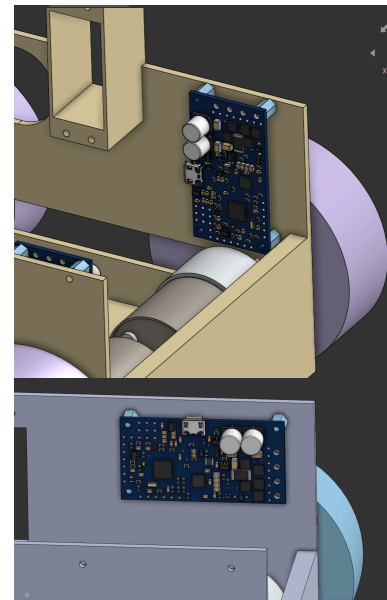


Figure 10: Changes to Motor controller location

4 Design Solution

This section presents the final design and implementation of the Sumobot system developed for the competition. It provides a detailed overview of the robot's mechanical, electrical, and control features. It also evaluates how the final design meets established Customer Requirements and Engineering Specifications. Key performance categories, including mobility, durability, and competitive effectiveness, are discussed to demonstrate the success of the final design.

Team Orange's final Sumobot design applies a torque-dominant drivetrain, a current-capable electrical architecture, and a manufacturable structural frame to meet defined customer requirements and engineering specifications. The drivetrain consists of four 12V DC motors with a 131:1 gear ratio, selected to maximize torque output for pushing interactions while maintaining controllable velocity within the competition ring. This configuration supports high traction and minimizes wheel slip during engagement due to the 4-wheel setup. The motors are controlled by two electronic speed controllers rated for 6.5V–30V input and up to 15A output current, providing sufficient current capacity to stall conditions without exceeding component limits. The controlling method is an RC transmitter and receiver system, enabling direct user input for maneuverability and engagement positioning. Electrical power is supplied by a 2200 mAh battery, selected to satisfy runtime requirements while maintaining adherence to the system weight constraints and ensuring stable voltage delivery under load. The structural system consists of a 3D printed PLA frame designed to meet the 20 cm × 20 cm footprint limitation and overall mass requirement of 3kg. The frame geometry supports component mounting, drivetrain alignment, and load transfer during pushing interactions. PLA was selected for its manufacturability and ability to support rapid iteration during the design process. The design solution prioritizes torque generation, electrical reliability, and structural integration, ensuring that all subsystems operate within specified limits while meeting customer expectations for competitive performance, controllability, and durability.

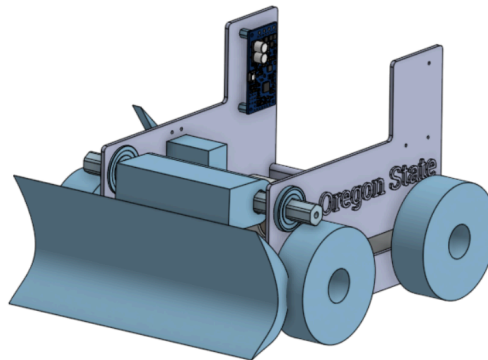


Fig. 11: Final Design Solution

4.1 Description of Solution

The implemented solution integrates the drivetrain, electrical system, and structural components into a unified system capable of consistent operation under competition conditions. The four-motor drivetrain distributes torque across four total contact points, increasing traction and improving stability during combat interactions. The electronic speed controllers regulate motor input, enabling controlled acceleration, deceleration, and directional changes while maintaining operation within rated current limits. The RC control system provides real-time command input, allowing the operator to adjust positioning and

engagement strategy based on opponent behavior. The PLA frame maintains structural integrity under loading conditions observed during testing and provides a stable platform for all subsystems. The primary offensive mechanism is a deployable ramp designed to engage the lower geometry of opposing robots. The ramp generates a lifting effect that reduces the normal force acting on the opponent's drive wheels, thereby decreasing available traction. This interaction increases the effectiveness of the drivetrain by improving the force balance during pushing. The mechanism operates without causing damage to opposing robots, satisfying competition rules related to non-destructive interaction. System validation included functional testing of drivetrain torque output, maneuverability, and ramp engagement performance. Results confirmed that the system met all engineering specifications, including dimensional constraints, electrical performance limits, and operational reliability. Repeated testing demonstrated stable controller performance under load, consistent motor response, and no structural failure of the PLA frame. The final system satisfies customer requirements for performance, control, durability, and effective engagement.

4.2 Project Results

Results from testing of our final design iteration included extensive testing of our 11 core Engineering Specifications (ES) [Figure 3], with only 8 of those ES's needing extensive testing. With several of our testings relying on being able to sustain and withstand at least 9.0 lbs of force acting on it, we performed this testing by attaching a 9 pound weight to the robot and observed how it would react when the weight was suspended over a table [Figure 12]. From that test, we observed that the robot managed to keep a steady location, but noticed that there were some slipping issues due to the low surface areas of the wheels only being in contact with the middle, which led to lesser friction resistance in the total design. Other functions of the robot relied on quick, small turning, as well as reliable acceleration and deceleration on command. Finally, our testing of the battery involved driving it in a simulated area to estimate battery life for competition. That test ran for 15 minutes, at which point testing was stopped as we would only need to be operational for 9.0 minutes for each match during competition, which is a successful test.

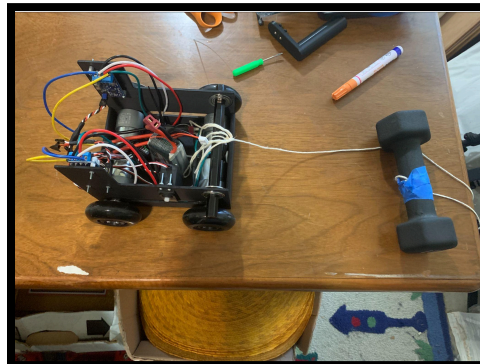


Figure 12: Force withstanding testing

Having worked on the development of the Sumbot for 5 months, Team Orange created a competitive RC controllable robot to compete in the Sumobot Competition at the Oregon State Engineering Expo this upcoming June. As the design has been discussed in the Design Solution, Team Orange hopes that the complexity of their robot will gain the attention of the Expo's participants and spark excitement in the audience when their robot competes in the Sumobot competition.

Aside from the performance based results from Team Orange's robot, every member in Team Orange has significantly grown from this experience of building a robot. Now, each member recognizes the amount of detail that goes into the designing something new and the degree of coordination that is necessary to

effectively build a robot as a team. This project overall challenged each member forcing them to develop and learn new skills that would be beneficial for any path they may choose after college.

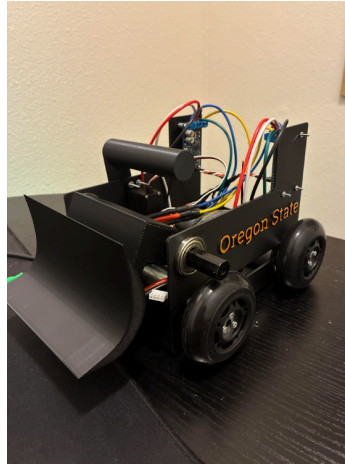


Figure 13: Final Design Physical Results

5 LOOKING FORWARD

Continuation of this project may include a variety of improvements that can change the outcome of the sumobot's performance. Factors such as manufacturing methods, materials used, structural optimization and stress analysis, and advanced electrical analysis would all be beneficial in the second iteration of this project.

Different manufacturing methods can allow the use of more suitable materials for a sumobot. Manufacturing methods such as sheet metal working or bending can allow the use of sheet aluminum, a lightweight material but significantly stronger than PLA. Should the next contestants in the sumobot competition have access to a metalworking shop, this technique and material could be used for the robot's frame/chassis, or offensive mechanism. Team Orange may also suggest an advanced analysis on stress concentrations and structural optimization. Using advanced techniques to predict the stress distribution on the sumobot during competition could allow for failure mode predictability. This predictability could allow for the structural support of problem areas and better identification of weak points. Lastly, advanced electrical analysis could be conducted to ensure the most efficient use of power supply and distribution. By using electrical optimization techniques, a team could fit the most adequate size/weight battery in the robot to acquire the greatest pushing force and runtime based on weight and power supply combined.

6 CONCLUSIONS

As previously stated, the general goal of this project is to build a robot to compete in the Sumobot competition hosted on June 5, 2026. This robot needs to initially fit inside a 20cm by 20cm square, weigh less than or equal to 3 kilograms, and be remote controlled by a driver during competition. The deeper goal of the project involved using the robot to promote STEM, gaining the interest of youth and guests during outreach events. This implies that the robot also has to be aesthetically pleasing and perform well during competition. The final result that Team Orange was able to produce was a 4 motor RC controlled robot that utilizes a deployable plowing mechanism to push opposing teams outside of the defined area in the game.

Alongside with the defined rules set in the competition that Team Orange had to adhere to, social and economic considerations were also accounted for when developing the robot. Because the intent is to operate the robot in a social environment, design decisions were made to make the robot as engaging as possible. Numerous testing procedures before the competition have been performed to ensure that the robot performs safely to the standards set in the House of Quality and entertains an appealing performance. As promotion of STEM to the community was another goal for this project, the team wanted to inspire groups and individuals to also design and build robots they are passionate about. Team Orange limited themselves to a realistic budget of \$1000 to set an example that capable robots can be made.

7 REFERENCES

[Include here all references cited, following the reference style described in the Style Guide. There should only be one Reference list in this report, so all individual section or subsection reference lists must be compiled here with the main report references. If you wish to include a bibliography, which lists not only references cited but other relevant literature, include it as an Appendix.]

Abstract References

Background References

Design Processes References

[1] Zaręba, Maciej, et al. "Mini-Sumo Robot Design with Respect to the Control System." *Problems of Mechatronics Armament Aviation Safety Engineering*, vol. 13, no. 3, 2022, pp. 131-144, <https://doi.org/10.5604/01.3001.0016.0056>.

[2] "What is a Battery Eliminator Circuit? How Does it Work?" *Best Technology*, Apr. 2025, www.bestpcbs.com/blog/2025/04/what-is-battery-eliminator-circuit-how-does-it-works/.

Design Proposal References

[3] RoboGames. (n.d.). *SumoBot Rules – All Robot Events Rules*. Retrieved January 17, 2026, from <https://robogames.net/rules/all-sumo.php>

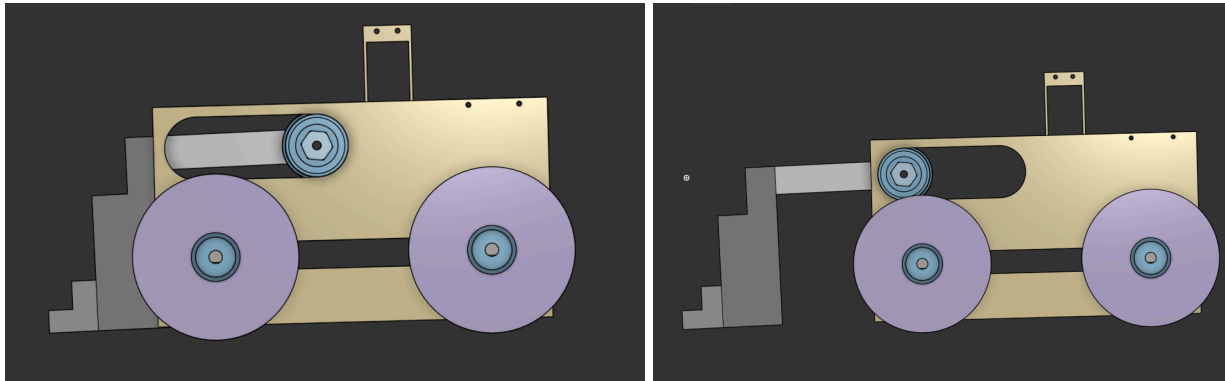
Design Solution References

Looking Forward References

Conclusion References

8 APPENDICES

8.1 *Appendix A: Offensive Mechanism Before/After Deployment Pictures*



This appendix is in order to show the before and after for our offensive mechanism deployment. As seen in the first image, at the beginning of a match, the mechanism would fall down so it is flush with the edge of the robot wall and its base is on the ground, from there, the servo would push it to its maximum extending length feature in the second image. Where springs attached to the shaft would bring it back to its natural position as soon as it had finished its linear motion.